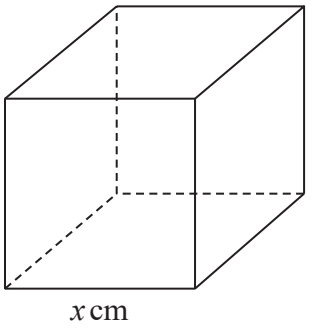
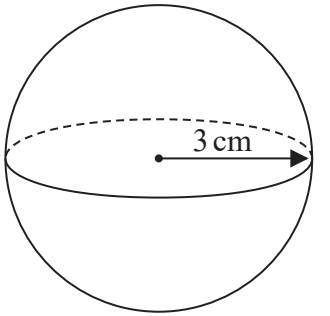
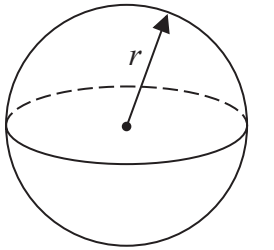
26 The diagram shows a cube with edges of length x cm and a sphere of radius 3 cm.







Surface area of sphere = $4\pi r^2$

The surface area of the cube is equal to the surface area of the sphere.

Show that $x = \sqrt{k\pi}$ where k is an integer.

Cube surface area = $6x^2$.

Sphere surface area = $4\pi(3^2) = 36\pi$.

Equate: $6x^2 = 36\pi \rightarrow x^2 = 6\pi$.

As $x > 0$, $x = \sqrt{(6\pi)}$, so $k = 6$.

4 marks total
$6x^2$
36π
Equate and divide
$x=\sqrt{(6\pi)}$, $k=6$

(Total for Question 26 is 4 marks)

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